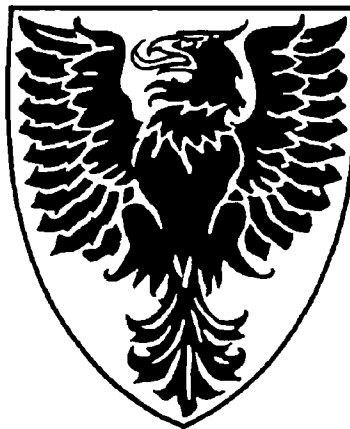


Seventeenth Annual In-House Convention

Department of Psychology
Dalhousie University

22-23 April 1991



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Schedule

Monday, 22 April

9:15-9:30 Opening Remarks: R. Brown

9:30-10:30 Session I. Chair: Jim Clark

- 1) E. Coscia
- 2) C. Beaver and D.E. Mitchell
- 3) W.K. Honig
- 4) D.P. Phillips and J. Kelly

COFFEE BREAK

10:50-12:15 Session II. Chair: Elizabeth Coscia

- 5) H. Abe, B. Rusak and H. Robertson
- 6) R.S. Rodger
- 7) V. Corkum
- 8) I. Smith and J. McGlone
- 9) B. Moore

LUNCH IN LOUNGE

12:30-1:30 Session III. Posters:

- 10) S.A. Clain, S.L. Hargrave, S.A. Sark and D.P. Phillips
- 11) C.L. Johnson, C.L. Ryan and R.E. Brown
- 12) J. McGlone
- 13) C.L. Ryan, R.E. Brown and A.M. West
- 14) S. Gustys
- 15) S. Hall and D. Mitchell

1:30-2:45 Session IV. Chair: Vin LoLordo

- 16) K. Rajashekhar and J.C. Wilkens
- 17) K. Frame and D. Williams
- 18) J. Barresi
- 19) Chin Lau and S. Nakajima

COFFEE BREAK

3:05-4:30 Session V. Chair: Bruno Losier

- 20) S.L. Wright
- 21) S. Shaw, K. Kokic and D. MacIntosh
- 22) J. Hamm and P. McMullen
- 23) D.P. Phillips and P. Tallal
- 24) A.J. Cohen, D. Dunphy, M. Lamoureux and B. Frankland

BEER IN LOUNGE

Tuesday, 23 April

9:15-10:30 Session VI. Chair: Amanda Pontefract

- 25) R. Brown, T. Murdoch, W. Moger and P. Murphy
- 26) M. Yoon and M.J. Yoon
- 27) P.C. Kind, C.J. Beaver and D.E. Mitchell
- 28) A. Kingstone and R. Klein
- 29) J.A. Goldman and J.C. Fentress

COFFEE BREAK

10:45-12:00 Session VII. Chair: Cathy Ryan

- 30) D. Colquhoun
- 31) K. Semba
- 32) P.D. Cole
- 33) E. Pyza
- 34) P.A. McMullen

LUNCH IN LOUNGE

12:15-1:30 Session VIII. Posters:

- 35) N. Akhtar and P.J. Dunham
- 36) V. Bolivar and J. Barresi
- 37) M. Farrell and R. Klein
- 38) N. Harvey
- 39) H. Schellinck and R.E. Brown
- 40) M. Patriquin

1:30-2:30 Session IX. Chair: Ray Klein

- 41) P.G. Ritvo, J.D. Fisk, C.J. Archibald and T.J. Murray
- 42) K.G. Bina, K. Semba and B. Rusak
- 43) S. Bishop
- 44) C. Moore

COFFEE BREAK

2:45-3:45 Session X. Chair: Jo Durup

- 45) J.H. Brandstätter, H.S. Seyan and I.A. Meinertzhagen
- 46) J. McNulty
- 47) B. Losier and K. Semba
- 48) B.W. Frankland

COFFEE BREAK

4:00-5:00 Session XI. Chair: Doug Williams

- 49) M. Ozier and A. Blumenthal
- 50) R. Croll
- 51) M. Lamoureux
- 52) J. Fentress, V. Bolivar, E. Coscia and W. Danilchuk

Beer and pizza will be served in the Lounge immediately after.

Abstracts

1. Elizabeth Coscia

The bum of the Banff bear and other grizzly tales

Wild timber wolves are slowly repopulating the Canadian Rockies. Recently, wolves have migrated into the southeast region of Banff National Park in Alberta. I collected radio-telemetry and observational data on members of one of the remote valley packs near their den and rendezvous sites from May - June, 1990. During this time of year, pack movements are less extensive to maintain close proximity with their young. Aside from encounters with wolves, I did have to contend with "Bruce" the bear and his "family". I will relate tales from this wilderness experience and speculate on a preliminary data analysis of the packs' homesite and rendezvous attendance.

2. C. Beaver and D.E. Mitchell

C-fos expression in the visual pathways of dark-reared kittens following initial exposure to light

The anatomical and physiological development of the central visual pathways of frontal eyed mammals is influenced strongly by early visual experience. In fact, the organization of central visual structures may remain close to the level that existed at birth in animals deprived of any visual input. However, upon subsequent exposure to light the anatomical and physiological organization of these structures is altered in a dramatic fashion. We have demonstrated that one of the family of immediate-early genes (c-fos) is expressed in visual cortical (but not subcortical) neural structures shortly after such animals are first exposed to light. This suggests that this gene and possibly other proto-oncogenes may play an important early role in mediating the anatomical changes that are induced by early visual experience.

3. W.K. Honig

Pigeons discriminate uniform from mixed arrays of visual elements

One property of an array of visual elements is the degree of uniformity -- the extent to which the elements are all the same, or mixed. Pigeons were trained to respond to uniform arrays of two kinds of elements, and not to respond to mixed arrays of the same elements. We obtained orderly gradients that reflected the degree of uniformity (or mixture). This discrimination is sensitive to the discriminability of the elements. It can be transferred between different sets of elements following appropriate training procedures.

4. Dennis P. Phillips and Jack B. Kelly

Physics and brain design, or what rats tell us about where god went wrong

By comparison with cats and monkeys, rats and other small mammals are very poor at localizing sounds in space. The most recent accounts of this disparity are couched in terms of the ethological demands on the species, or in terms of the correlation between localization acuity and the horizontal width of the species' visual streaks or areae centrali (since this width determines the auditory localization acuity required to bring the target into the part of the visual field with greatest acuity). These accounts in no way explain why small rodents are so poor at localizing sounds: they simply point to correlates of that performance. Jack and I measured the ability of rat cortical auditory neurons to encode interaural time differences, which are a major cue for sound location. We found that the dynamic range of this code was the same as that seen in cats and primates; this despite the fact that rats have much smaller heads, and therefore will only ever encounter very small interaural time disparities. We suggest, then, that for rats, there exists a mismatch between the behaviorally-relevant range of interaural time differences and the range of time differences that the animal's brain encodes. This may be one reason why small mammals are so poor at sound localization.

5. Hiroshi Abe, Benjamin Rusak and Harry Robertson

Photic induction of Fos protein in the suprachiasmatic nucleus is inhibited by NMDA receptor antagonists

Exposure of hamsters to light can induce c-fos immunoreactivity in cells of the suprachiasmatic nucleus (SCN), which is a dominant pacemaker in the mammalian circadian system. Excitatory amino acids (EAAs) have been implicated in both the entrainment pathway by which light synchronizes circadian rhythms and in the induction of c-fos in other parts of the nervous system. We examined the effects of pre-treatment with MK-801 and ketamine which are non-competitive N-methyl-D-aspartate (NMDA) receptor antagonists on the induction of Fos-like immunoreactivity (Fos-lir) in cells in the hamster SCN. Pretreatment with these drugs blocked photic induction of Fos-lir in a dose-dependent manner in the rostral SCN and ventral portions of the caudal SCN, but not in a discrete region of the dorsolateral SCN. These results suggest that the NMDA-type of EAA receptor may be involved in the photic induction of Fos-lir in most of the SCN but not in one portion of the SCN.

6. R.S. Rodger

Betcha Bob's $E\beta$ is better than the bruited β by a bundle

The traditional anova procedures is badly flawed so far as type-1 error-rate is concerned, leading to a dramatic loss of power β as the numerator degrees of freedom of the variance-ratio increases, given a fixed $N\Sigma(\mu_j - \mu)^2 / \sigma^2$. There are also serious ambiguities involved in setting β for a pre-chosen $\Sigma(\mu_j - \mu)^2$. These problems are resolved if Rodger's (1967-) method is used. That sets rejection rates for null contrasts at $E\alpha$ when they are true and at $E\beta$ when they are false by the pre-chosen amount $g \sqrt{\Sigma c_j^2}$. The amount of treatment effect claimed after analysis can be justified by maximum likelihood if $E\beta$ has been chosen wisely.

$$E\beta = .90$$

7. Valerie Corkum

Development of early interlimb coordination

Recently, increasing attention has been devoted to the application of dynamical theories of motor coordination to infant motor development. A number of studies indicate little synchrony in early movement patterns but a significant increase over the first 6 months. A recent study by Thelen and colleagues found the legs to be coupled in the form of compensatory limb movements at 6 weeks of age. However, as yet, the developmental trend of compensatory limb movements has neither been documented for the legs nor has its existence even been explored in the arms. The present study looks developmentally at these two types of motor coordination (i.e., synchronous and compensatory movements) in both arms and legs in order to understand better the nature and ontogeny of early interlimb coordination.

8. Isabel Smith and Jeannette McGlone

Sodium Amytal memory testing: Use in prediction of post-surgical amnesia

The intracarotid sodium amytal procedure is used worldwide to determine lateralization of speech (Wada, 1949) and memory (Milner et al., 1962) prior to temporal lobe resection for epileptic seizure control. Jack et al. (1988) have advocated injection of sodium amytal directly into the posterior cerebral artery (PCA), rather than the internal carotid artery (ICA), for the selective examination of memory functions. They argue that more direct inactivation of the posterior hippocampus would improve prediction of post-surgical amnesia; the increased risk associated with catheterizing the PCA makes this a controversial suggestion. In addition, Milner (1972) and Rausch et al. (1989) claim that using the standard ICA injection, filling of the PCA is not necessary to produce amnesia. We examine the relationship between filling of the PCA (blind angiographic ratings) and recognition memory in 83 epileptic patients undergoing the traditional ICA procedure. We also discuss other factors related to the reliability and validity of the intracarotid sodium amytal memory test.

9. Bruce Moore

The evolution of avian imitation*

1. The ability to imitate movements, previously demonstrated clearly only among "higher" mammals (certain primates and cetaceans), is shown in a psittacine bird. The animal, a Grey Parrot (*Psittacus erithacus*) uses its feet, wings, head, beak and tongue in unambiguous imitation of human movements.
2. Several lines of evidence suggest that the bird's imitation is not homologous to that in mammals. The process therefore seems to have evolved at least twice.
3. A "new" form of imitative learning, *non-vocal mimicry*, in which bodily movements are used to create mimetic sounds, is tentatively identified in passerine and psittacine birds.
4. Hierarchical and phylogenetic relationships suggest that the evolutionary path of avian imitation has been: Song/call learning --vocal mimicry -- non-vocal mimicry -- movement imitation.

*Parts of this talk previously presented in Psychology Proseminar

footers —

10. S.-A. Clain, S.L. Hargrave, S.A. Sark and D.P. Phillips

Studies on the human pure-tone audiogram

The normal human audiogram has a U-shape, and it depicts the relative sensitivity of the listener to pure tones of frequencies spread across the audible range. It is thought to represent the envelope of the cochlear/neural channels with the greatest sensitivities. We conducted three perceptual experiments to study these channels. First, we obtained audiograms from three normal listeners, and confirmed the U-shaped distribution of audiometric sensitivity. Second, we studied the effects of continuous wide-band noise masking on the audiogram. These studies revealed that, for any given noise level, thresholds for tone detection were most elevated for frequencies to which the subject was most sensitive in quiet. This differential sensitivity means that the sensory channels comprising the audiogram can function independently. Third, using a modern pure-tone masking paradigm, we explored the frequency selectivity of individual sensory channels. The resulting psychophysical "tuning curves" were strikingly reminiscent of the frequency selectivity of cochlear nerve fibers seen in neurophysiological studies. Taken together, these psychophysical studies are compatible with the view that cochlear output consists of narrowly-tuned channels that function independently.

11. Carrie L. Johnson, Catherine L. Ryan and Richard E. Brown

Measures of reactivity and CNS development in the hooded rat pup

Lewis, Worobey, and Thomas (New Directions for Child Development, 45, 33-37) report a relationship between early measures of reactivity and later health status in human neonates. Reactiveness of the neonate at 2 days of age is predictive of the robustness of the 18 month old infant. We will present data illustrating the relationship between measures of reactivity (ultrasonic vocalizations, von Frey hair reaction, and open field testing) and measures of robustness (weight and neurobehavioural maturation) in the Long-Evans rat pups. The aim of this study is to determine the predictive value of these early measures of reactivity in the rodent for later health status. This could then be used as a model for screening environmental factors for human teratogenic potential.

12. Jeannette McGlone

Memory complaints before and after temporal lobectomy

This clinical study identified clinical outcome variables that influenced memory complaints in surgical candidates with temporal lobe epilepsy. Right-handed adult epileptics (N = 47) with medically intractable seizures were assessed both pre-operatively and one year after a left or right temporal lobectomy. The Memory Observation Questionnaire measured self-report of memory complaints. The delayed Paired Associates Test and the Rey-Osterrieth Figure measured verbal and nonverbal memory. The results showed that memory complaints diminished post-operatively, but this decline was less striking in patients of average or higher WAIS-R Full Scale IQ. Regression analyses revealed that dosage of anti-epileptic medications, seizure frequency and depression were predictive of post-operative memory complaints. Objective memory scores deteriorated after the surgery depending on side of the resection and sex of the patient. Together these findings suggested that most epileptic patients hold a positive impression that their memory function improved after temporal lobectomy, an impression that appears to be influenced by positive surgical outcomes rather than actual performance.

13. C.L. Ryan, R.E. Brown and A.M. West

Is the measurement of isolation cries a sufficient indicator of distress?

It has been proposed that neonatal rodent ultrasonic vocalizations (UVs) can be used to screen drugs for particular psychopharmacological profiles and human teratological potential. Previous literature proposes that an increase in UV rate should occur following administration of a drug that is stressful, while a decrease in UV rate characterizes a drug with anxiolytic potential.

We will present data which challenges this interpretation. We have found that the prototypic anxiogenic compounds lithium chloride (LiCl) and corticotropin releasing factor (CRF) fail to increase the rate of neonatal vocalizations as would be expected from the standard experimental interpretation. Rather, our data suggests that in evaluating the psychopharmacological profiles of drugs in the neonate, a change in UV rate alone, may be the significant factor and that the direction of this change cannot, by itself, predict the affective properties of the drug.

14. Stephanie Gustys

Effects of exposure to a map on accuracy of distance estimation

In a study of cognitive mapping, 30 Dalhousie University freshmen first drew a map of their campus. Two-thirds of the subjects then briefly studied a printed map of the campus, and one-third studied a map of St. Mary's campus. Then, all subjects estimated distances between locations on the Dalhousie Campus. Of the subjects who had previewed the Dalhousie map, one-half were asked to remember that map, while the other half was asked to remember their drawn map. It was hypothesized that exposure to the Dalhousie map combined with instructions to consult this map would lead to the greatest sensitivity in the estimation task. Although mean slopes and correlations followed this pattern, the differences were not significant. There were however significant differences in the kind of imagery used in the distance estimation task: the group who had not previewed the Dalhousie map reported significantly greater use of imagery of locomotion and less use of a map-like image. These data are consistent with the notion that multiple representations of large-scale space can be independently consulted in spatial problem solving.

15. Susan Hall and Donald Mitchell

Things that go jump in the dark

Compared vernier and step displacement acuities in human foveal vision. When test and reference stimuli were of the same contrast, vernier thresholds were lower than step thresholds. For opposite-contrast stimuli, vernier thresholds increased four-five fold but step thresholds by very little. Further evidence was found demonstrating the difference between spatial and temporal visual mechanisms. Preliminary data obtained in a cat subject are also shown.

16. Rajashekhar K.P. and J.L. Wilkens

Wonder if smoking releases dopamine in crabs too?

Perfusion of nicotine into isolated thoracic ganglion of crab, caused an increase in the frequency of bursts from the ventilatory neural pattern generator (CPG). Similar action was not elicited by acetylcholine nor was it blocked by curare. Dopamine elevated the burst frequency of the CPG. Domperidone, a D2 specific dopamine antagonist had a reversible depressant effect, whereas, SKF-83566. HCl a D1 specific antagonist had no effect. The excitatory action of nicotine could be blocked by Domperidone but not SKF-83566. HCl. Dopamine appears necessary for the 'generation and maintenance of the rhythm of the CPG. Nicotine is known to release dopamine from dopaminergic neurons in higher vertebrates and may be acting via a similar mechanism in crabs too.

17. Kelly Frame and Doug Williams

The shocking truth: Fear-evoking stimuli acquire their properties independently of one another

It has been popular in the fear conditioning literature to assume that predictive stimuli compete with each other to signal a fear-eliciting event. We provide "shocking" new evidence that this claim is untrue under many conditions: Contextual and temporal predictors of footshock evoke the same level of anticipatory fear in rats (in the form of conditioned freezing) regardless of whether they have supposedly been made redundant by the presence of a more predictive discrete tone signal.

18. John Barresi

About the 'aboutness' of intentionality's intentions': Can we find a meaning for 'meaning' in the causes of 'because's'?

Over the past few years I have been approaching the concept of intentionality from a variety of directions, both in my own independent work and in collaborative work with others including Valerie Bolivar, Tim Juckes and Chris Moore. Yet, despite an increase in my understanding of the nature of intentionality, I still find it a somewhat mystical concept. In the present paper I will try to de-mystify intentionality by focusing on its major defining attribute - 'aboutness' - and by trying to provide a causal-materialistic reduction of this attribute. Even if unsuccessful, it is my hope that this attempt will - at least - put some greater light on the meaning of intentionality, which I believe is a concept of central importance for the ultimate integration of the bio-psycho-social sciences.

19. Chin Lau and S. Nakajima

Interaction between D1 and D2 dopaminergic receptors in self-stimulation of the lateral hypothalamus in the rat

Previous studies in our laboratory have shown that the reinforcing effect of brain stimulation can be attenuated by either selective D1 dopamine antagonist or a selective D2 antagonist, and enhanced by D2 agonists but not by a D1 agonist. Rats with electrodes implanted into the lateral hypothalamic area were trained to press a bar for a train of pulses with variable frequencies (12 - 200 pulses per s) and a frequency-response curve was plotted. A D2 agonist quinpirole (2.0 mg/kg ip) increased responding when injected alone, but failed to show this effect when injected after a D1 antagonist SCH 23390 (0.04 mg/kg, ip). SCH 23390 shifted the frequency-response curve to a higher frequency range regardless of whether it was injected alone or after quinpirole. These results suggest that the enhancement of reinforcing effect by a D2 agonist manifests itself as long as D1 receptors are activated by dopamine normally present in the rat brain but not when they are blocked by a D1 antagonist. These findings are consistent with the reports that these subtypes of dopamine receptors work synergistically.

20. Sandra L. Wright

Neuromotor maturation and the ontogeny of ultrasonic vocalizations in the California mouse (*peromyscus californicus*)

Ten female and eleven male *P. californicus* pups from 8 litters were individually marked, weighed and tested for ultrasonic vocalizations (UVs). UVs, locomotion, surface righting, negative geotaxis, acoustic startle reflex and eye opening were recorded from 2 to 30 days of age. UVs at 37, 42, 47 and 52 kHz were digitized and recorded by computer and an event recorder program which was simultaneously used to record locomotor behaviour. Results were analyzed by sex of pup and litter size. Put weight was negatively correlated with UVs ($r = -.58$, $p < .05$). UVs decreased as pups aged ($p < .01$). Females vocalized more than males ($p < .01$) and vocalizations increased for both males and females after they spent more than 180 s in coordinated movement ($p < .01$). There was no effect of litter size on UVs. After the auditory startle reflex was shown, UVs decreased significantly ($p < .01$). Although there was no effect of eye opening on UVs, pups from litters of two opened their eyes earlier than pups from larger litters ($p < .01$). Males opened their eyes earlier than females ($p < .05$). These results indicate that *P. californicus* has a pattern of UV development that is related to weight, age, sex, development of coordinated movement and auditory startle reflex. Surface righting, negative geotaxis and eye opening were not related to the development of UVs.

21. Steve Shaw, Kathy Kokic and Don MacIntosh

Ears on insect knees

Crickets carry a pair of directionally-selective hearing organs on their forelegs, which they use to home in on the calling songs of potential mates. They are distantly related to cockroaches, which have a simpler organ that until recently was believed to respond only to vibration, but which we showed last year also responds sensitively to sound. The roach subgenual ("below the knee") organ is particularly interesting in that it forms a conceptual bridge to explain the evolution of the more elaborate, pure auditory cricket's ear from an ancestral pure vibration receptor. We have found, unexpectedly, that the roach ear is also strongly directional. According to the standard physical theory applied to insect hearing organs, possession of directionality implies the presence of two sound input pathways to the organ. In crickets, the two entry points are an external tympanum on the leg, and internal air tubes (tracheae) connected to openings (spiracles) on the sides of the body. Only the tracheal pathway is obvious in roaches, with no evident tympanum. Attempts to pinpoint how and where sound gets into the knee will be described, along with tests that demonstrate that roaches can respond to sound, albeit feebly, given an appropriate behavioral assay.

22. Jeffrey Hamm and Paddy McMullen

Identification, starting at the top

Viewing the same object from different angles results in different retinal images. We are able to recognize these different visual inputs as the same object. This is known as object constancy. When line drawings of familiar objects are presented to subjects, the time to name the object increases linearly for objects rotated up to 120 degrees from the vertical. This function suggests that during object identification, the input is being normalized to the upright, until a match is found in memory. Biederman (1989) has suggested that the effects of orientation are due to finding the top of the object. In order to test this hypothesis, valid and invalid precues of object orientation will be presented prior to the display of the object to be named. If the effect of orientation is in finding the top, as Biederman has suggested then the valid cue condition should show less of an effect of orientation than the invalid cue.

23. Dennis Phillips and Paula Tallal

Temporal auditory processing deficits in developmental dysphasia

The most comprehensive, if somewhat controversial, account of developmental dysphasia is Tallal's. Briefly, she argues that the speech reception deficit is due to an auditory temporal processing disorder: the disordered mechanism is thought to be cortical in locus, and to affect the perception of both linguistic and nonlinguistic sounds. The most compelling demonstration of the impairment lies in the inability of these children to discriminate the order of two brief (75 ms) tone pulses if the duration of the silent period between them is less than 100 to 300 ms, a task which normals perform with near-perfect accuracy for inter-stimulus intervals as small as 0 ms. The intended connotation of the term "temporal processing" is, however, far from clear in Tallal's published work. By the same token, nothing is known about the generality of any deficit across auditory perceptual dimensions that have their origins in different forms of time-related or time-dependent sound processing. In this talk, I will present sketches of two of a long series of experiments, based on modern auditory neuroscience, that we have planned to address these lacunae.

24. A.J. Cohen, D. Dunphy, M. Lamoureux and B. Frankland

Battery includes music

Musical stimuli engage a variety of sensory, perceptual and cognitive processes. Yet, none of the numerous neuropsychological tests in existence examines these processes. Drawing upon tasks used in research in the psychology of music, we devised such a test. It examines discrimination, memory, frequency direction, tonal structure, meaning, and preference. Data from two sets of college students are presented showing high agreement on test items. The potential value to neuropsychology is discussed.

25. Richard Brown, Thad Murdoch, Will Moger and Paul Murphy

Hormonal and experiential factors regulating paternal behaviour in male rodents

Our current knowledge of the hormonal regulation of parental behaviour in rodents is based primarily on the maternal behaviour of the rat. Because the rat is a polygynous species, in which the male shows little parental care, the rat model may not be appropriate for the study of the hormones and paternal behaviour in biparental species such as gerbils and California mice. We are developing a model of the hormonal and experiential factors involved in the stimulation of male parental care in biparental rodents. I will outline this model and describe the results of three experiments on hormones and male parental behaviour based on this model: (1) changes in testosterone and prolactin levels across the reproductive cycle in male rats and gerbils; (2) hormonal responses of male and female rats and gerbils to one hour of pup replacement following four hours of pup removal; (3) the effects of prolactin on the parental behaviour of virgin male gerbils.

26. Myong Yoon and Myonghae Joy Yoon

Children's reasoning for prediction of human actions: Developmental studies

We study young (2-5 years old) children's cognitive development with respect to their abilities to make appropriate predictions of a protagonist's actions and to explain their own judgements. Four brief movies are designed to test (1) "Simple Desire Reasoning", (2) "Belief/Desire Reasoning", (3) "Discrepant Belief Reasoning", and (4) "False Belief Reasoning" (Wellman & Woolley, 1990; Perner, 1990) as follows: [a] Let a child (the subject) assume an active role of 'explainer' who tells his/her addressee (the experimenter) what is going on in a test story enacted in the movie that the child has been watching (instead of being passive observer of the test situation presented by the experimenter). Then the experimenter asks the child to predict what the protagonists in the movie will do. [b] Video-recording of all utterances and communicative gestures produced by the children and the experimenter during the entire discursive organizations as well as for their lexico-syntactic structures and semantic cohesion. These analyses will reveal not only the cognitive abilities of the children for different types of reasoning but also their pragmatic competence in communicating their thoughts and emotions at various stages of their development. [c] Longitudinal studies with individual children (2 to 5 years old) to denote the individual patterns of developmental changes through consecutive tests.

27. P.C. Kind, C.J. Beaver and D.E. Mitchell

The effects of monocular deprivation and reverse occlusion on the expression of CAT-301 in the LGN of kittens

Recent attempts to reverse the severe behavioural effects of an early period of monocular deprivation (MD) in kittens by subsequent imposition of a period of reverse occlusion (RO) have revealed a very diverse range of eventual outcomes from good vision to both eyes to either monocular or binocular amblyopia. We have examined the expression of CAT-301 (a monoclonal antibody that recognizes a proteoglycan associated with the surface of Y-cells in the cat visual pathway) in the visual pathways of animals subjected to certain of these rearing regimens and have found that many regimens of RO, and especially those that do not favour one eye, lead to a substantial diminution of CAT-301 immunoreactivity in all laminae of the LGN.

28. Alan Kingstone and Ray Klein

Attention and express saccades

Our initial studies confirm the observation that removing a fixation stimulus shortly before the appearance of a visual target elicits unusually rapid ("express") saccadic responses even when target direction is uncertain. The popular explanation for express saccades is that attention must be disengaged before a saccade can be initiated, and that offset of any attended stimulus (pre-) disengages attention, thus permitting express saccades. Studies directly assessing the relationship between attention and express saccades will be presented.

29. J.A. Goldman and J.C. Fentress

Does the squeaking wolf always get the response?

Research at the Dalhousie Animal Behaviour Field Station (DABFS) is currently examining communication between a mother timber wolf and her pups, during the first 5 postnatal weeks. Previous work on activity inside the den, has shown that although the mother rarely responds to her pups' vocalizations with overt behaviours, she does monitor their vocal output by exhibiting subtle changes in ear direction. Now, the pups' responsiveness to the mother's vocalizations is under investigation. A 460-hour videotape database permits examination of vocal and behavioral activity inside the den, as well as, vocalizations which occur outside the den. Pup behaviour is coded for response to the adult 'Squeak', a common social vocalization. Preliminary data suggests that not all 'Squeaks' elicit the same response.

A

30. David Colquhoun

Neuropsychological assessment of subcortical dementia in Huntington's Disease

Individuals with Huntington's Disease and Dementia of Alzheimer's Type (DAT) were examined on a neuropsychological battery designed to assess frontal, memory, and language abilities. The purpose of testing both DAT and HD subjects was to examine the distinctiveness of cortical and subcortical dementias and, by doing so, clarify the possible role that the basal ganglia pathology has in the mediation diverse set of cognitive processes. This comparison is especially important in light of the lack of direct comparisons made between HD and DAT. It was hypothesized that deficits in HD would be characterized by focal frontal deficits and subtle declines in memory while DAT would be marked by language difficulties coupled with generalized memory deficits. Preliminary analysis indicates the existence of qualitative differences between HD and DAT which appear to reflect divergent pathological processes. It is proposed that cognitive changes in HD and DAT support the position that dementia may be differentiated in terms of a cortical/subcortical dimension.

31. Kazue Semba

What does the shape of neurons tell you about what neurons do?: Morphology of cholinergic and non-cholinergic cortically projecting neurons in the rat basal forebrain

Neurons appear in different shapes. Whatever factors might be involved for these various shapes to form during development and then change during normal aging (as well as in pathological conditions), neuronal morphology often provides interesting insights as to functions of individual neurons. Previously we have shown that there are two physiological subtypes of cortically projecting neurons in the basal forebrain, and that these are also morphologically distinct, and have hypothesized that one subgroup is cholinergic and the other non-cholinergic. In the present study, we test the morphological aspect of this hypothesis, using retrograde tracing, single cell injections, and immunohistochemistry.

32. P.D. Cole

Pigeon umwelt: Is a picture worth a thousand words?

Operant chamber techniques have yielded a diversity of information regarding various facets of pigeon vision. More recently complex photographic material in the form of slides, depicting sundry categories such as trees, cats, cars and people, has been employed in category discrimination tasks. In such studies the assumption is presumably made that the photographs are perceived by the Pigeons as representative of the three-dimensional objects or scenes that they depict. Are there enough salient dimensions present in a slide photograph to allow it to function as a suitable representation of the Pigeons environment and, if so, could it be used to explore the complexity of the Pigeon Umwelt? Some recent work in this area will be presented.

33. Elzbieta Pyza

The effect of quipazine - a serotonin agonist, on the circadian rhythm of locomotor activity and the serotonin level in the brain of the house cricket (*Acheta domestica* L.)

The rhythm of locomotor activity in insects results from the function of an endogenous pacemaker system and entrainment of this system by environmental factors (light) changing during the 24 hour period. The aim of this study was to examine the role of serotonin (5-HT) in the regulation of circadian rhythms. Results obtained indicated that the circadian system controlling locomotor activity rhythms in the house cricket is composed of two oscillators which act jointly (in DD - constant dark), or separately (in LL constant light) and that 5-HT takes part in the regulation of overt rhythmicity. In LL, injections of quipazine affected the phase of two components of the rhythm or changed one of the components present in activity into the second one (changes of the rhythm period). In DD, quipazine administrations caused only phase shifts of the rhythm which were phase-dependent. Quipazine injections decreased also the 5-HT concentration in the cricket brain which may be a result of a negative feedback process triggered by direct or indirect activation of 5-HT receptors.

34. Patricia A. McMullen

Symmetry and orientation normalization in the recognition of naturalistic stimuli

Tarr and Pinker (Psychological Science, 1990) showed that mental rotation is used in recognizing misoriented asymmetrical patterns, but not symmetrical patterns, using novel stimuli. They inferred that object-centered representations code spatial relations among features on one dimension only. The generality of this finding was assessed by retrospectively analysing three experiments' worth of latencies to name misoriented drawings of common objects. For the first block of trials, symmetrical and asymmetrical objects showed equivalent effects of orientation between upright and 120°. In contrast, effects of orientation were significantly greater for asymmetrical than symmetrical objects over three subsequent blocks of trials. These results are consistent with the hypothesis that object-centered representations code object features along a single dimension.

35. Nameera Akhtar and Philip J. Dunham

Comprehension strategies of young children

Previous research (Shatz, 1978) has indicated that young children are biased to respond to language with action. This study attempted to replicate this finding in a group of 64 24 month olds. These children heard 30 prerecorded sentences differing in syntax presented through a speaker in a robot (thereby eliminating extralinguistic cues such as gesture and gaze). Their responses were videotaped and coded as Action, Informing, or Other. Results indicated that the children were not biased to respond with action, and were able to discriminate among the five sentence types. The data suggest that this discrimination may be based on surface markers such as the presence of an action verb, and the mention of an object present in the immediate context.

36. Valerie Bolivar and John Barresi

The secret to creating life

Do humans equate spontaneous movement with being alive and having internal states? Do children think the same way as adults on these issues? Recently, researchers have been grappling with the effects of movement patterns on animacy and intentionality attributions. Unfortunately, most of these researchers have used objects familiar to the subject and hence they are subject to biases associated with previous experience. In the present study, we examine the role of endogenous movement in the attribution of animacy, motivation and internal states to unfamiliar objects. Subjects in grades 1, 4 and 7 and university were asked to respond to the following statement "Tell me a story about what you have just seen", after viewing a computer generated display consisting of two circular stimuli exhibiting three types of movement. Results indicate that age, gender, as well as movement patterns all contribute to differences in these attributions.

37. Mark Farrell and Raymond M. Klein

The effect of regional compactness upon region discrimination time is modulated by oculomotor programming

Podgorny & Shepard (1983) demonstrated that reaction time to determine if a target is on or off an attended region decreased with increasing compactness of that region. Six experiments explored the nature of this compactness effect and its relationship to visual attention. Subjects viewed a 3 x 3 grid in which there were shaded and unshaded regions. In four experiments their task was to determine a) if a target letter appearing in the shaded or unshaded region was a T or L, or b) if a target stimulus was presented in the shaded or unshaded region. Eye position monitoring was used to ensure that subjects were not shifting gaze to inspect the display. The compactness effect was not obtained in any of these experiments. In two further experiments eye position monitoring was NOT employed and the pattern reported by Podgorny & Shepard - a negative correlation between RT and compactness - was observed. The RT/compactness relationship discovered by Podgorny & Shepard appears to be mediated by the programming of the oculomotor system and does not reflect the distribution of endogenously controlled attention.

38. Natasha Harvey

Integration of kinematic information in emotional judgments

Adults rated the happiness/sadness of 36 short film excerpts of a person whose facial and body movements varied from excerpt to excerpt. There were three different facial expressions (sad, neutral and happy as reflected by mouth position and eye focus) and three different body movement patterns (sad, neutral and happy as reflected by position and rate of limb movement) combined factorially. The influence of both sources of information on ratings was highly significant and in the manner predicted. Thus, emotional meaning can be communicated by multidimensional kinematic information. Results are compared with those from preschool children.

39. Heather Schellinck and Richard E. Brown

Discrimination of human male odours by rats is not influenced by the relatedness of the urine donors

Rats can be trained in an olfactometer to discriminate between the urine odours of MHC congenic rats and mice. They learn these discriminations quickly and remember the information from one daily session to another. Such results indicate that constant odour cues are used to discriminate between these urines (Schellinck, Brown and Slotnick, *Anim. Learn. Behav.*, in press). To determine if rats can discriminate between human urines, we tested five Long-Evans rats in a go/no discrimination in an olfactometer. Discriminable differences were present in urines from monozyotic twins, dizygotic twins, siblings and unrelated males. Analysis of the errors to criterion over eight 200 trial sessions indicated that the four human urine odour discriminations were equally difficult to learn. Although all test rats were able to discriminate the urine pairs to a criterion of 90% correct within one or more of the eight sessions, only one of the rats was able to maintain this level of performance from session to session. The other four subjects either relearned the discrimination within a session or performed at chance levels on subsequent sessions. When the ability of rats to learn to discriminate between the urine odours of rats, mice and humans is compared, it appears that constant cues are being used to discriminate between the urine odours of rats and mice whereas this is not the case for human urine odours.

40. Maria Patriquin

The effects of play structures on infant initiated imitative play

Nonverbal social exchanges or interactions are crucial steps in infants becoming full partners in a social communicative context. Studies in the area have shown that imitation is a key element in the development of coordinated joint attention. It is argued that the period in which infants are sensitive to the imitative structures is at 24 month of age. The present study observed 30 nine month olds to investigate whether infants are sensitive to the imitative structures of play at this age. Subjects were randomly assigned to three treatment groups, Synchronous-Contingent, Asynchronous-Contingent and Noncontingent. Infants in the Synchronous-Contingent group engaged in more imitative play while the infants in the other two treatment groups engaged in more parallel play. Being imitated actually encouraged the infants to engage in more reciprocal imitative play. Although not statistically different, the Noncontingent infants displayed a tendency to engage in less of both imitative and parallel play. Noncontingent infants were said to be somewhat sensitive to their play structure given that they looked more at the experimenter prior to responding than the Synchronous-Contingent group did.

41. P.G. Ritvo, J.D. Fisk, C.J. Archibald and T.J. Murray

Neurological and psychosocial predictors of mental health in patients with Multiple Sclerosis

In the initial study of an ongoing project investigating quality of life and psychosocial functioning in MS patients, 89 subjects from the Dalhousie MS Research Clinic were assessed. Scores on the Mental Health Inventory were used as the dependent measure in a multiple regression analysis that included measures of social support, fatigue, responsiveness to stress, physical handicap, neurological disability, duration of disease and disease activity. Results indicate that social support and fatigue are important issues for MS patients. These two factors alone accounted for 44.8% of variance in Mental Health scores of the total sample. Furthermore, standard status measures of neurological disability and physical handicap were not significant predictors of Mental Health status in MS patients.

42. K.G. Bina, K. Semba and B. Rusak

Localization of cholinergic and nerve growth factor-receptor neurons projecting to the suprachiasmatic nucleus of the rat

Circadian rhythms in mammals are regulated by neural mechanisms centered in the suprachiasmatic nucleus (SCN) of the anterior hypothalamus. Recent immunohistochemical findings indicate that both choline acetyltransferase (ChAT), the synthesizing enzyme for acetylcholine, and nerve growth factor-receptor (NGF-R) are present in the SCN. In order to identify the source of this immunoreactivity, a combination of retrograde tracing and immunohistochemistry for ChAT and NGF-R was performed. Following iontophoretic injections of fluorogold into the SCN, in addition to previously described areas, retrogradely labelled neurons were seen in the substantia innominata, the part of the magnocellular basal nuclear complex near the SCN. Some of these retrogradely labelled cells were also single- or double-labelled with ChAT and NGF-R. Unilateral knife cuts directed lateral to the SCN decreased NGF-R immunoreactivity in the ipsilateral SCN, whereas, unilateral enucleation decreased immunoreactivity in the contralateral SCN.

43. Scott Bishop

The relationship between catastrophizing and negative affect

Two studies were conducted to examine the cognitive-affective correlates of catastrophizing. Catastrophizing has been defined as a tendency to engage in excessively negative cognitions related to aversive stimuli such as pain. Study 1 examined the reliability and construct validity of a self-report measure of catastrophizing cognitions. A 13 item scale, developed for this study was shown to have high internal consistency ($\alpha = .90$). In addition, the scale showed the same relationship with subcomponents of depressive symptomatology shown with clinical measures of catastrophizing. Study 2 examined subjects evaluations of the valiance of positive, negative and pain-related words following a success or failure experience. Results indicated that catastrophizers who experienced success on a problem solving task rated positive and pain-related words more negatively than noncatastrophizers. Study 2 also addressed the test-retest reliability of the catastrophizing measure. The results are discussed in terms of the relationship between catastrophizing and negative affect.

44. Chris Moore

Imagination and social understanding

It has been claimed that social understanding is achieved not so much through the development and application of a "theory of mind" but through the ability to pretend or imagine self in situations appropriate to another time or person. In this paper, I discuss recent work of ours on social reasoning tasks, in which children's judgements have to be made either from a third-person perspective or from an imagined first-person perspective. Initial results seem to support the idea that imagined or pretend contexts facilitate social reasoning.

45. J.H. Brandstätter, H.S. Seyan and I.A. Meinertzhagen

The dynamic properties of synapses in the fly's lamina

In the first optic neuropile, or lamina, beneath the fly's (*Musca domestica* L.) compound eye photoreceptor terminals (R1-R6) and interneurons are reciprocally connected. At each afferent synaptic sites a presynaptic ribbon in R-R6 abuts a postsynaptic tetrad containing processes of two monopolar cells, L1 & L2. L2 in turn provides feedback input to R1-R6 at a synaptic dyad. We have examined both afferent and feedback synapses, in each case following the loss of their main target cells. When R1-R6 degenerated by photoablation, presynaptic ribbons of feedback synapses separated from the presynaptic membrane and soon appeared free floating in L2's cytoplasm. Despite the presence of floating ribbons the number of L2 synapses increased twofold during an overlapping period of reactive synaptogenesis. When L1 and L2 degenerated by lesioning, presynaptic ribbons of afferent synapses were only moderately affected with slight decreases in their frequency. Thus many presynaptic ribbons in R1-R6 outlive their postsynaptic partners. This difference in the reaction of photoreceptor and monopolar cells following the loss of postsynaptic partners, reflects a greater dependence of monopolar cells upon their targets, R1-R6, than of R1-R6 upon their targets, L2 and L1.

46. John McNulty

Psychological sequelae of carbon monoxide poisoning and their treatment with hyperbaric oxygen

One of the most frequently observed symptoms of carbon monoxide (CO) poisoning is "mental" or "cognitive" impairment. These symptoms often persist even when other medical and neurological symptoms have disappeared. The standard treatment for coexposure is hyperbaric oxygen. If symptoms are minor, or exposure time is short, patients are sometimes treated only with oxygen at normal atmospheric pressure or are not treated at all. There are, however, "hints" that failure to treat even mild cases may result in relatively permanent neuropsychological deficits. Some preliminary data and suggestions for research on the problem will be presented.

47. Bruno J. Losier and Kazue Semba

Cholinergic and aminergic inputs from the mesopontine area to the basal forebrain and thalamus: A hypothesis for brainstem involvement in cortical arousal

Researchers of behavioral states have investigated the role of specific direct and indirect cholinergic inputs of the basal forebrain and mesopontine tegmental cholinergic areas in the regulation of waking and sleep. This study examines the contribution of tegmental cholinergic, serotonergic and dopaminergic cell groups in modulating the activity of both thalamic and basal forebrain neurons in cortical arousal. Following multiple injections of both FITC beads into the basal forebrain, and Fluorogold into the thalamus, alternate sections from the forebrain and brainstem were processed for immunohistochemistry with antibodies to visualize cholinergic and aminergic neurons. Injection sites were plotted and alternate brainstem sections were surveyed for the presence of triple labeled neurons.

48. B.W. Frankland

Non-temporal and temporal effects on the perception of pattern in sequential stimuli

The analysis of the perception and/or cognition of pattern in sequentially presented stimuli is constrained by non-temporal and temporal factors that limit the number and type of possible temporal patterns. This work will discuss the effects of such constraints on the perception of pattern in the contour (the pattern of up and down) in sequentially presented tones. In particular, a brief review of the role of tonality (essentially, a non-temporal constraint) and the role of short term memory (essentially a temporal constraint) on the perception of pattern in contour will be presented.

49. Marcia Ozier and Andrea Blumenthal

Mood and memory: Open your Ears and close your eyes

Usually we remember better if given a category cue to the meaning of the word we are seeking in memory than if we are given an alphabetic one. That finding reverses if we are induced into depression before trying to remember. Induced depression results in a dramatic decrease in the potency of category cues to meaningful words. Previous reports have been based upon an experimental design in which words were presented visually and test cues presented visually. The current experiment asked whether the decrease in potency of category cues under induced depression would occur if presentation and test were auditory rather than visual. The loss of meaning as a function of induced depression seems to be specific to the visual modality. The moral is: If you feel down, open your ears and close your eyes. This remedy is not recommended if you feel down while you are driving a car.

50. Roger P. Croll

Injury-induced neuroplasticity in the snail

Following a crush to a central pathway in the snail, the nervous system responds by regenerating lesioned axons, by rerouting some fibres to contralateral pathways and by sprouting new projections from uninjured neurons. Together these responses lead to supernumerary neurites throughout the nervous system. These aberrant projections are gradually lost, and normal projection patterns return by 5 weeks after the lesion. When regeneration is prevented by cutting rather than crushing the nerve, supernumerary elements continue to increase with no return to normal numbers evident after 15 weeks. Together these results suggest that the cessation of sprouting and the retraction of supernumerary fibres are dependent upon successful regeneration of the injured neurons.

51. Michael Lamoureux

Studies of ambiguous pitch: Direction and reaction time measures

In line with our interest in the representation of sequential patterns, listeners judged the pitch direction of pairs of randomly selected Shepard tones (synthesized from 10 consecutive octave components with amplitude envelope peaking over central components) from the set of 12 semitones in the octave. Direction judgments were consistent for short absolute distances between tones and were bivalent for distances of 5-6 semitones. Reaction time data and the pattern of directions preceding ambiguous intervals reveals whether ambiguous pitch is disambiguated by preceding redundant directional patterns.

52. John Fentress, Valerie Bolivar, Elizabeth Coscia and Wanda Danilchuk

The marvels of mutant mice in motion

The marvels of mouse movements provide the inspiration for this project. The small but sophisticated brains of these little beasts enable them to articulate beautifully coordinated actions. These, as other animal (and human) actions, challenge us to think carefully about how mammalian brains achieve their essential balance between isolable operations and the rules of combination in space and time among these operations. We shall treat the troops with our first attempts to employ benign developmental analyses of neurological mutant strains to examine fundamental problems of pattern formation in behavioral organization. The data are preliminary, but you'll love our mice!

